REINFORCEMENT LEARNING 2026Z · PROJECT PRESENTATION 1

MiniHack: Wand of Death

One reinforcement-learning agent for all four levels of a NetHack task: find the wand, deal with the minotaur, reach the goal.

 Player  Wand of death  Minotaur  Staircase

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Team no. 5

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Interactive presentation and playable game

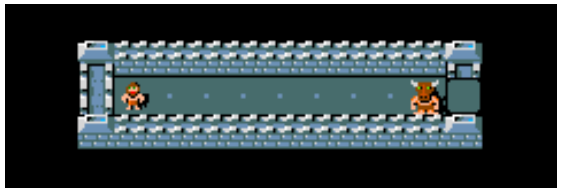
wojtke.github.io/minihack-wand-agent

Environment and task

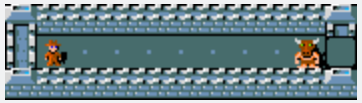
MiniHack [1] (built on the NetHack Learning Environment): a turn-based dungeon. The agent must use a wand of death against a minotaur. Single agent, one opponent.

Easy

Action limit: **50**



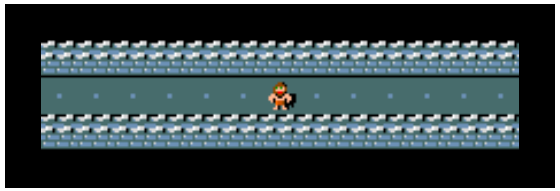
Wand in inventory; sleeping minotaur in a corridor.



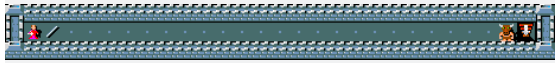
Goal: kill the minotaur.

Medium

Action limit: **150**



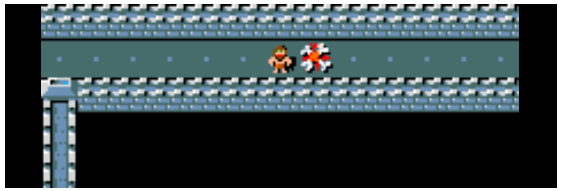
Wand on the floor; sleeping minotaur blocks the corridor.



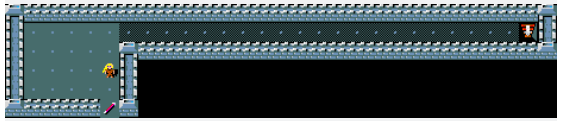
Goal: reach the staircase.

Hard

Action limit: **400**



Find the wand in the start room; the minotaur is awake.



Goal: reach the staircase.

Pro

Action limit: **1,000**



Wand hidden somewhere in a maze; minotaur awake.



Goal: reach the staircase.

Top: frames of recorded episodes (videos in the interactive version). Bottom: full maps revealed for illustration only; the agent sees only explored terrain.

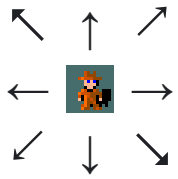
[1] M. Samvelyan et al., “MiniHack the Planet”, NeurIPS 2021 Datasets and Benchmarks Track, arXiv:2109.13202.

Actions, observations, reward

An episode ends on success, death or the action limit.

Action space

Discrete, 11 native inputs (Restricted set)



8 directions
move or attack

3 commands: pick up · use (zap) a wand · fire

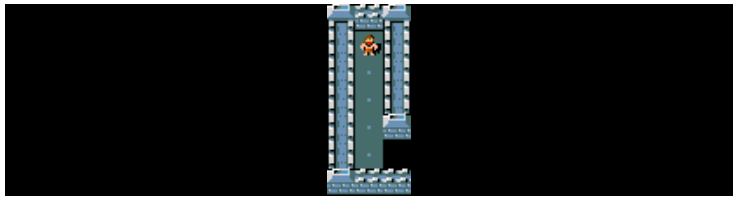
Using a wand = 3 decisions

1. “Use a wand”
2. select the wand from inventory
3. choose a direction

Prompts reuse the same inputs: a direction key can aim instead of moving.

Observations

Partial observability: visible objects and remembered terrain; unexplored areas stay hidden.



Pro at the start: only a short corridor is visible.

Map
21×79 arrays of glyphs, characters, colours

Inventory
item names and letters

Character state
health, position, game time

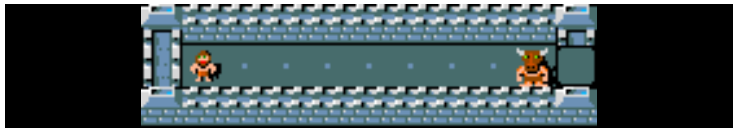
Messages and prompts
e.g. “In what direction?”

Reward (default)

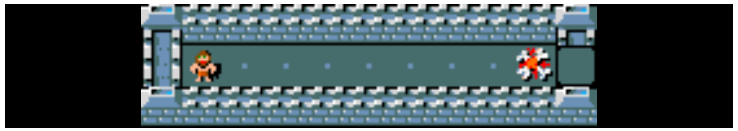
+1	**Success:** minotaur killed (Easy) / staircase reached (other levels)
0	**Failure:** death or action limit
−0.01	**Step penalty** when game time does not advance (menu reply, blocked move)

Easy: a successful wand attack

Before · minotaur alive



After · minotaur killed · **+1**



Sparse: practically one non-zero reward per episode.

Configuring the environment

Four difficulty levels, each available with two action sets; further parameters of MiniHack/NLE.

Difficulty levels

LEVEL	LIMIT	WAND	MINOTAUR	VARIES BETWEEN EPISODES
Easy	50	in inventory	asleep	nothing
Medium	150	on the floor	asleep	nothing
Hard	400	in start room	awake	player and wand position
Pro	1,000	in a maze	awake	wand position

Terrain is fixed per level. Success: kill the minotaur (Easy) or reach the staircase (others).

Agents and opponents

Single-agent environment, **one opponent** (minotaur). Optional: random extra monsters (`spawn_monsters`), a pet (`pet`), custom levels via MiniHack's level generator / des-files.

Parameters

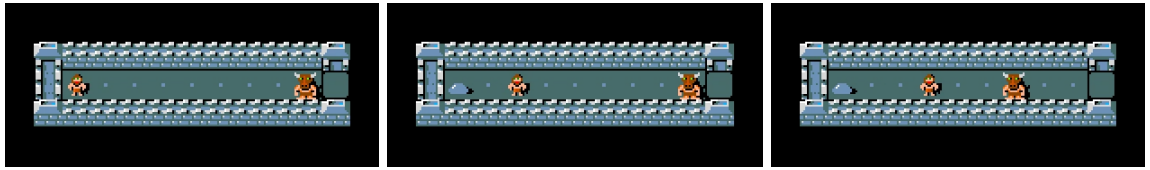
Action set	Restricted: 11 inputs · Full: 85 inputs (`…-Restricted-v0` / `…-Full-v0`)
Action limit	`max_episode_steps` , counts every input including menu replies
Observations	`observation_keys` : glyphs, chars, colours, stats, message, inventory, pixels, 9×9 crops around the player
Rewards	`reward_win` , `reward_lose` , `penalty_step` , `penalty_time` , `penalty_mode`
Seeds	control placements in Hard/Pro; separate ranges for training, validation, test

> **Our setup:** Restricted actions, default limits and rewards. Training only: +0.01 per newly visited square (max +2 per episode), removed for evaluation.

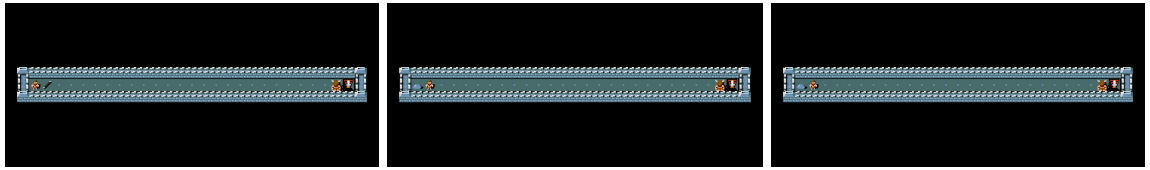
A random agent

Each step: one of the 11 inputs chosen uniformly at random. One recording per level, seed fixed in advance. Frames: start · middle · end.

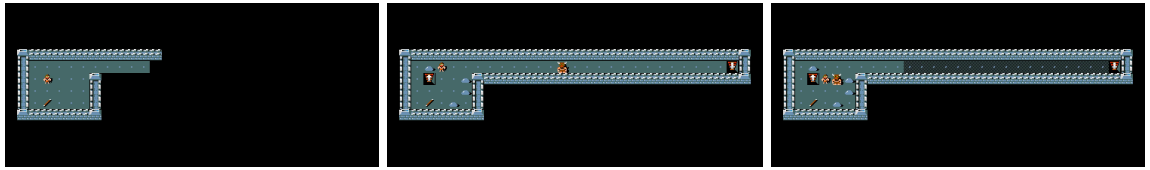
Easy — action limit after 50 actions



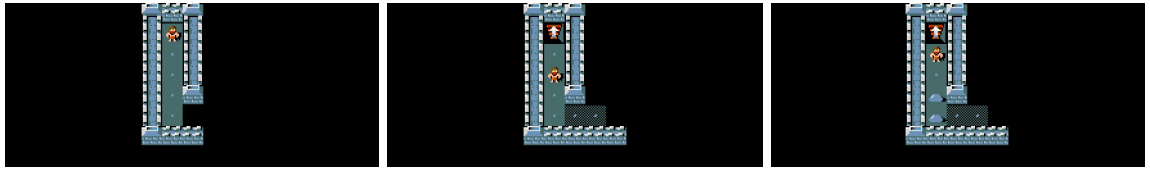
Medium — dies after 111 actions



Hard — dies after 32 actions



Pro — first 150 of 1,000 actions



500 GAMES PER LEVEL	EASY	MEDIUM	HARD	PRO
Success	**28** (5.6%)	**0**	**0**	**0**
Death	320	186	500	10
Action limit	152	314	0	490

Videos: slide 3 of the interactive presentation. Easy succeeds only when random inputs happen to form the wand sequence aimed at the minotaur. With the Full set of 85 inputs: no successes in 100 games per level.

Plan

One agent for all four levels, trained with reinforcement learning.

Starting point	**Double DQN**, MLP 128×128, replay buffer, target network, ε-greedy. Input: 463 features from public observations (local map crop, monsters, remembered items and exits, inventory, health, prompt). One network for all levels, no level label.
Final goal	The same agent also wins on Pro, where the wand has to be found in a maze. Ideas: memory (recurrent policy), example games from a scripted player.
Baseline	The random agent (previous slide) and the published IMPALA results [1]; later, our first Double DQN as the reference for every new method.
Comparisons	With / without exploration bonus · several training seeds · memory and demonstrations for Pro.

Intermediate goal	One agent that solves Easy, Medium and Hard.
Expected performance	Easy and Medium close to 100%, Hard around 80%, some wins on Pro, i.e. beyond the published results [1] below.
Metrics	Reward only says win or loss, so we also track: win rate on unseen seeds · how games end (minotaur, own ray, action limit) · wand picked up · squares explored · actions per win.
Evaluation	Greedy policy, native reward only, game seeds never used in training.

Published results [1]: IMPALA, mean episode return, 5 runs

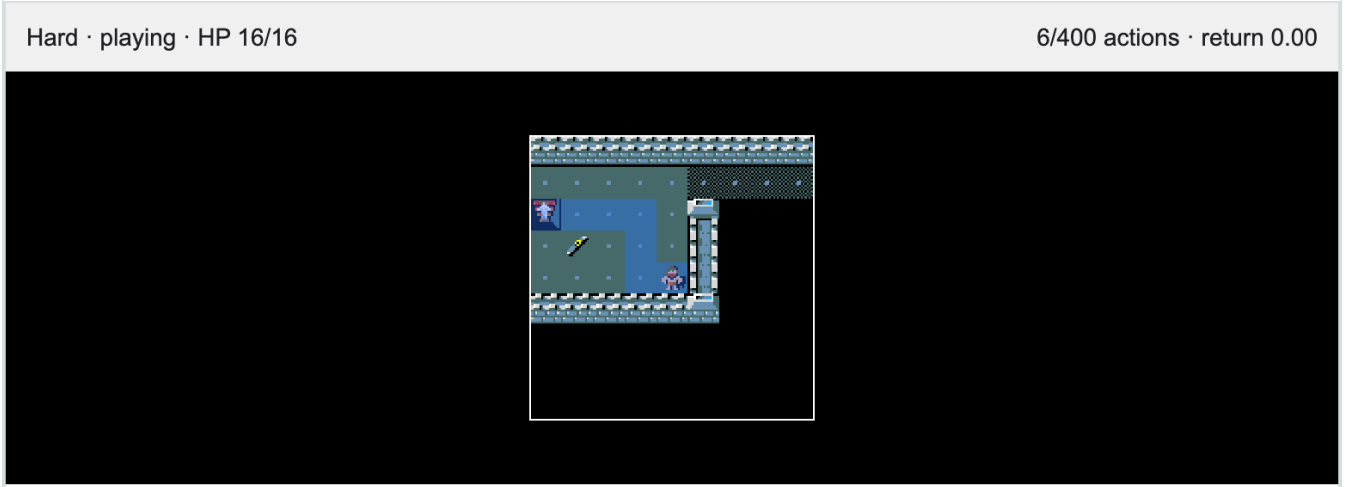
WOD-EASY	WOD-MEDIUM	WOD-HARD	WOD-PRO
≈ **0.95** within 25M steps	**0** after 100M steps	**0** after 250M steps	not reported

“None of the runs is able to make progress on WoD-Medium […] due to insufficient state-action space exploration.” [1]

[1] M. Samvelyan et al., “MiniHack the Planet: A Sandbox for Open-Ended Reinforcement Learning Research”, NeurIPS 2021 Datasets and Benchmarks Track, arXiv:2109.13202 (Fig. 8, Fig. 31).

Interactive demonstration

The original NLE engine compiled to WebAssembly runs entirely in the browser, with MiniHack's task rules and native inputs.



Agent view, centred on the player: the 9×9 window the network receives; blue squares are where the agent has stood. Outside it, the agent knows only where the nearest **items**, **staircases** and **monsters** are; far ones are shown at the edge, in their direction.

- All four levels are playable by a human, using the same 11 inputs as the agent.
- **Agent view**: what the network receives, i.e. a 9×9 window around the player (blue = visited squares) plus positions of the nearest items, staircases and monsters.
- Controls: WASD + Q/E/Z/C move or aim, P picks up, R uses the wand, F fires.

Try it:
wojtke.github.io/minihack-wand-agent/#5